



**Kavayitri Bahinabai Chaudhari North Maharashtra University,
Jalgaon
Faculty of Science and Technology**



Syllabus

BCA (Honors/Research) Third Year SEMESTER – V and VI

Semester-wise Code structure for B.C.A. (Honors/Research)
Program as per NEP 2020, for Affiliated Colleges w.e.f – June 2024.

BCA (Honors/Research) – First Year, SEMESTER – I, Level – 4.5

Course	Course Type	Course Code	Course Title	Credits	Teaching Hours/ Week			Marks (Total 100)			
					T	P	Total	Internal (CA)		External (UA)	
								T	P	T	P
DSC-1	DSC	CA-111	Essential of Computers and Programming	2	2	--	2	20	--	30	--
DSC-2	DSC	CA-112	Programming using C++	2	2	--	2	20	--	30	--
DSC-3	DSC	CA-113	Practical based on Programming using C++	2	--	4	4	--	20	--	30
OE-1	OE	-	BCA students shall opt OE from basket	2	2	--	2	20	--	30	--
OE-2		-	BCA students shall opt OE from basket	2		2	2				
SEC-1	SEC	CA-114	Office Management Tools	2	2	--	2	20	--	30	--
VSC-1	VSE	CA-115	Web Design-I	2	2		2	20		30	
AEC-1	AEC	--	English -1	2	2	--	2	20	--	30	--
VEC-1	VEC	--	Environmental Science	2	2	--	2	20	--	30	--
IKS	IKS	--		2	2	--	2	20	--	30	--
CC-1	CC	--	Select any one from Basket of CC	2	2	--	2	50	--		--

BCA (Honors/Research) – First Year, SEMESTER – II, Level – 4.5

DSC-4	DSC	CA-121	Object Oriented Programming in C++	2	2	--	2	20	--	30	--
DSC-5	DSC	CA-122	Introduction to Operating System	2	2	--	2	20	--	30	--
DSC-6	DSC	CA-123	Practical based on CA-121 & CA-122	2	--	4	4	--	20	--	30
MIN-1	MIN	CA-124(A)	Basics of Management	2	2	--	2	20	--	30	--
		CA-124(B)	Basic Mathematics for Computing								
OE-3	OE	--	BCA students shall opt OE from basket	2	2	--	2	20	--	30	--
OE-4	OE	--	BCA students shall opt OE from basket	2	2		2	20		30	
VSC-2	VSC	CA-125	Web Design-II	2	2	--	2	20	--	30	--
SEC-2	SEC	CA-126	Introduction to Graphics Design	2	2	--	2	20	--	30	--
AEC-2	AEC	--	English -2	2	2	--	2	20	--	30	--
VEC-2	VEC	--	Constitution of India	2	2	--	2	20	--	30	--
CC-2	CC	--	Select any one from Basket of CC	2	2	--	2	50	--		--

Cumulative Credits For First Year – 44

Semester-wise Code structure for B.C.A. (Honors/Research)
Program as per NEP 2020, for Affiliated Colleges w.e.f – June 2024.

BCA (Honors/Research) – Second Year, SEMESTER – III, Level – 5.0

Course	Course Type	Course Code	Course Title	Credits	Teaching Hours/ Week			Marks (Total 100)			
					T	P	Total	Internal (CA)		External (UA)	
								T	P	T	P
DSC-7	DSC	CA-211	Data Structures	2	2	--	2	20	--	30	--
DSC-8	DSC	CA-212	Python Programming	2	2	--	2	20	--	30	--
DSC-9	DSC	CA-213	Practical based on Data Structures	2	--	4	4	--	20	--	30
DSC-10	DSC	CA-214	Practical based on Python Programming	2	--	4	4	--	20	--	30
MIN-2	MIN	CA-215(A)	Fundamentals of Accounting	2	2	--	2	20	--	30	--
		CA-215(B)	Graph Theory								
MIN-3	MIN	CA-216(A)	Practical on Tally	2	--	4	4	--	20	--	30
		CA-216(B)	Practical on Graph Theory								
OE-5	OE	-	BCA students shall opt OE from basket	2	2	--	2	20	--	30	--
VSC-3	VSC	CA-217	Ethical Hacking	2	2	--	2	20	--	30	--
CC-3	CC	-	Select any one from Basket of CC	2	2	--	2	50	--		--
FP-1	FP	CA-218	Field Project	2		8	8		20		30
AEC-3	AEC	MIL	Modern Indian Languages	2	2		2	20		30	

BCA (Honors/Research) – Second Year, SEMESTER – IV, Level – 5.0

DSC-11	DSC	CA-221	Database Management System	2	2	--	2	20	--	30	--
DSC-12	DSC	CA-222	Artificial Intelligence	2	2	--	2	20	--	30	--
DSC-13	DSC	CA-223	Practical based on Database Management System	2	--	4	4	--	20	--	30
DSC-14	DSC	CA-224	Practical based on Artificial Intelligence using Python	2	--	4	4	--	20	--	30
MIN-4	MIN	CA-225(A)	Enterprise Resource Planning	2	2	--	2	20	--	30	--
		CA-225(B)	Basics of Statistics and Probability for Computer Science								
MIN-5	MIN	CA-226(A)	Practical based on ERP using case studies	2	--	4	4	--	20	--	30
		CA-226(B)	Practical on Basics of Statistics and Probability for Computer Science								
OE-6	OE	-	BCA students shall opt OE from basket	2	2	--	2	20	--	30	--
SEC-3	SEC	CA-227	Advanced Graphics Design	2	2		2	20	--	30	--
CEP	CEP	-	Community Engagement and Service	2		4			20		30

		CA-326 (C)	Machine Learning									
DSE-4	DSE	CA-327(A)	Practical based Web Technologies-II	2	--	4	4	--	20	--	30	
		CA-327(B)	Practical based on Data Analytics II	2	--	4	4	--	20	--	30	
		CA-327(C)	Practical based on Machine Learning	2	--	4	4	--	20	--	30	
MIN-08	MIN	CA-328(A)	Social Media Marketing	2	2	--	2	20	--	30	--	
		CA-328(B)	Linear Algebra -II									
MIN-09	MIN	CA-329(A)	Practical based Social Media Marketing	2		4	4		20		30	
		CA-329(B)	Practical on Linear Algebra -II			4	4		20		30	
#OJT/ Int	OJT/Int	CA-330	On Job Training / Internship	4	--	8	8	--	40	--	60	
# Students need to complete one month on job training (OJT) or internship in any industry related to major subject.												

Course Code: CA-311
Course Title: Software Engineering

Course Code: CA-311	Course Category: DSC-15
Course Title: Software Engineering	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand what is software and the process in development of software 2. To understand various models and requirements needed in developing software. 3. To elaborate the concepts of software designing. 4. To apply testing techniques for software quality assurance.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Describe the System Development Life Cycle (SDLC) and its phases	2
2. Understand data collection techniques for requirement gathering in a software project..	2
3. Construct the E-R Diagram, DFD, Data dictionary, Decision tree about software	3
4. Describe software testing and quality assurance	2

Course Content:

Unit I: Software Concept and SDLC

(07 L, 10M)

- Definition of Software.
- Characteristics of Software.
- Types of Software- System, Application, Embedded, Web application A.I.
- Software Process Framework
- Software Engineering -Concept
- Introduction of Software Development life Cycle - SDLC.
- Waterfall Model,
- Prototyping Model,
- Spiral Model,
- RAD Model.

Unit II: Software Requirements Analysis and Specification

(09 L, 15M)

- Data and Fact Gathering Techniques:
- Interviews and Questionnaires,
- Group Discussion,
- On-site Observation,
- Review of Written Documents
- Roles of System Analyst: As an Architect, Change Agent, Investigator & Monitor, Organizer, Motivator
- Feasibility Study: Introduction,
- Types of Feasibility: Technical, Economical, Operational
- Software Requirement Specification (SRS)- Introduction
- Need and Importance of SRS

Unit III: Software Design Engineering

(09 L, 15M)

- Principles of Software Design
- Translating the analysis model into a software design
- Fundamental design concept-Abstraction, Refinement, Modularity
- Entity Relationship Diagram
- Data Flow Diagrams, DFD Levels
- Flowchart
- User - Interface design:
- Graphical interfaces,
- Elements of Good Interface design
- Designing application system using DFD, ERD
 - Library Management System
 - Hospital Management System

Unit IV: Software Testing & Quality Assurance

(05 L, 10M)

- Overview of Software Testing:
- Introduction to software testing
- Software Testing Principles
- Types of software testing - Black box testing, White box testing
- AI Tools for software testing: Jira
- Quality Assurance :
 - Software Quality - Garvin's quality dimensions, Mc Calls quality factors, ISO 9125 quality factors
 - Elements of Software Quality Assurance ISO 9000 & Certification

Reference Books:

1. **“Software Engineering a Practitioners Approach”**, by Roger S. Pressman, ISBN13: 9780071267823, 7th edition, McGraw Hill International Edition.
2. **“Fundamental of Software Engineering”**, by Rajib Mall, ISBN- 978-81-203- 3819 7 3 RD Edition, PHI Learning Private Limited.
3. **“Software Engineering”**, by Gaurav Gupta and Deepika Gupta”, ISBN:978 3843310130, Lambert Publishing pvt. Ltd
4. **“Software Engineering”**, by Udit Agrawal ISBN: 978-9350142363

Course Code: CA-312
Course Title: Programming using C#.Net

Course Code: CA-312	Course Category: DSC - 16
Course Title: Programming using C#.Net	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand the fundamentals of .NET platform and C# programming language. 2. To implement structured programming and object-oriented programming concepts using C#. 3. To analyze collections and exception handling mechanisms in C#. 4. To develop logical and problem-solving skills required for software development and placement preparation.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Understand .NET platform architecture and the structure of C# programs.	2
2. Implement control structures, methods, and exception handling in C#.	3
3. Apply collections and basic database connectivity using ADO.NET in C# applications.	3
4. Develop basic GUI applications using event-driven programming concepts in Windows Forms.	3

Course Content:

Unit 1: Introduction to .NET and C# Fundamentals

(06 L, 10 M)

- Introduction to .NET Framework and .NET Core.
- Structure of C# program – namespace, class, Main method.
- Compilation and execution process, data types and variables, type casting
- Operators – arithmetic, relational, logical, assignment.
- Console input and output, conditional statements, looping statements.

Unit 2: Object-Oriented Programming in C#

(08 L, 15 M)

- Introduction to object-oriented programming concepts, advantages of OOP
- Classes and objects, Encapsulation using access specifiers, properties (get and set), Inheritance.
- Constructors and its types, constructor overloading, methods and parameter passing.
- Polymorphism, method overloading, method overriding.
- Abstraction using abstract class, interface and its implementation.

Unit 3: Collections, Exception Handling

(06 L, 15 M)

- Arrays and their limitations, Introduction to Collections framework.
- List<T> collection – dynamic data handling and common operations (Add, Remove, Count).
- Comparison between Array and List (conceptual).
- Exception handling – try, catch, finally blocks, Multiple catch blocks and basic exception types.

Unit 4: Event-Driven Programming and GUI Concepts Database Connectivity

(10 L, 10 M)

- Introduction to event-driven programming, Difference between console and GUI applications.
- Introduction to Windows Forms and form creation.
- Common GUI controls – Label, TextBox, Button, RadioButton, CheckBox, ComboBox, ListBox, DataGridView.

- Event handling – Click, TextChanged, CheckedChanged events.
- Form input handling and basic validation techniques.
- Introduction to ADO.NET architecture.
- Database connectivity using SQL Server.
- SqlConnection, connection string concept, and SqlCommand for basic Insert, Update, Delete, Select

Reference Books:

1. **Mark J. Price** — C# 10 and .NET 6 – Modern Cross-Platform Development Fundamentals, Packt Publishing, ISBN-13: 9781801077361
2. **Andrew Troelsen & Phil Japikse** — Pro C# 10 with .NET 6, Apress Publications, ISBN-13: 9781484278680
3. **John Sharp** — Microsoft Visual C# Step by Step (9th Edition), Microsoft Press, ISBN-13: 9781509301041
4. **Herbert Schildt** — C# 4.0: The Complete Reference, McGraw-Hill Education, ISBN-13: 9780071741163
5. **Mahesh Chand** — ADO.NET Programming with C#, Microsoft Developer Network (Reference Edition)
6. **Karli Watson et al.** — Beginning Visual C# and Windows Forms Programming, Wrox Publications, ISBN-13: 9780764549484
7. **RB Whitaker** — The C# Player's Guide (5th Edition), Starbound Software, ISBN-13: 9780985580155

Course Code: CA-313
Course Title: JAVA Programming

Course Code: CA- 313	Course Category: DSC-17
Course Title: JAVA Programming	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand fundamental concepts and structure of Java programming. 2. To explain and apply object-oriented programming concepts in Java. 3. To design and develop GUI-based applications using AWT and Swing. 4. To understand process of database connectivity.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Describe Java architecture and programming fundamentals.	2
2. Develop object-oriented programs using classes, inheritance, and interfaces	3
3. Build GUI applications using Swing component	3
4. Apply database connectivity	3

Course Content:

UNIT 1: Java Fundamentals and Programming Basics

(6L, 12M)

- History and Features of Java
- JDK, JRE, JVM, JIT
- Compiling and Running Java Programs (Command Line).
- Command Line Arguments.
- Data Types, Variables
- Accepting Input using: BufferedReader, Scanner
- Operators and Control Structures (if, switch, loops)
- Arrays (1D & 2D)

UNIT 2: Object-Oriented Programming Concepts

(9L, 13M)

- Introduction to Classes and Objects
- Access Specifiers
- Constructors: Default, Parameterized.
- Polymorphism: Method Overloading & Method Overriding
- Inheritance and Types
- keywords: super, this, final
- Abstract Class and Interfaces
- Wrapper Classes
- Creating Packages
- Exception Handling : Types of Exceptions, Try, catch ,Finally.

UNIT 3: Graphical User Interface Programming using AWT & Swing

(8L, 12M)

- Introduction to AWT and Swing
- Difference between AWT and Swing.
- Layout Managers: Flow Layout, Border Layout, Grid Layout.
- Event Handling and Listeners.
- Swing components : JFrame, JPanel, JLabel, JTextField, JButton, JCheckBox, JRadioButton, JList.

UNIT 4: : Database Programming

(7L, 13M)

- Introduction to SQL Command: DDL, DML.
- Introduction to JDBC: Architecture (2-tier, 3-tier)
- JDBC Drivers
- Connectivity with Database: basic steps
- JDBC statement: Statement, PreparedStatement, CallableStatement
- JDBC ResultSet and types .

References:

1. **Herbert Scheldt** by Java 2: The Complete Reference.
2. **E. Balagurusamy** by Programming with Java, A primer.
3. **Cay S. Horstmann, Gary Cornell, Prentice Hall, Sun Microsystems Press** by Core Java Volume-I-Fundamentals.
4. **R. Nageswara Rao**- Complete reference Java Craig Walls-Spring in Action.
5. **Ben Piper**-AWS Certified Solutions Architect Official Study Guide.
6. **Maydene Fisher et al., Addison-Wesley** :JDBC API Tutorial and Reference —.

Course Code: CA-314
Course Title: Practical based on C#.Net

Course Code: CA-314	Course Category: DSC - 18
Course Title: Practical based on C#.Net	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand the fundamentals of the .NET platform and C# programming structure. 2. To apply programming constructs such as data types, operators, conditional statements, and loops to develop console applications. 3. To implement object-oriented programming concepts including classes, inheritance, polymorphism, and encapsulation in C#. 4. To utilize collections, exception handling, and basic database connectivity concepts using ADO.NET. 5. To develop basic Windows Form applications using GUI controls and event-driven programming concepts.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Understand the .NET platform and the structure of C# programs.	2
2. Develop console-based applications using data types, operators, conditional statements, and looping constructs.	3
3. Design and implement applications using object-oriented programming concepts such as classes, inheritance, polymorphism, and abstraction.	3
4. Apply collections, exception handling, and basic database connectivity using ADO.NET in C# applications.	3
5. Develop basic event-driven Windows Form applications using GUI controls.	3

Course Content:

Assignments

Assignment 1: Basic C# Program and Console Output

Write a C# program to demonstrate basic program structure (namespace, class, Main method) and display student details such as name, roll number, and course using console output statements.

Assignment 2: Data Types and Operators

Create a console application to accept two numbers from the user and perform arithmetic operations (addition, subtraction, multiplication, division) using appropriate data types and operators.

Assignment 3: Student Marks and Grade Calculation

Develop a console application to accept marks of multiple subjects, calculate total and percentage, and display grade using conditional statements (if-else).

Assignment 4: Loop-Based Application

Write a C# program to generate a multiplication table or calculate factorial of a given number using looping statements (for / while / foreach).

Assignment 5: Employee Class using OOP Concepts

Create an Employee class with properties (EmployeeId, Name, Salary), constructors, and a method to display employee details, demonstrating class, object, encapsulation, and constructor concepts.

Assignment 6: Inheritance and Polymorphism

Develop a program using Employee (base class) and Manager (derived class) to demonstrate inheritance and method overriding for salary calculation.

Assignment 7: Student Record Management using List

Create a console application to store student records (Roll No, Name, Marks) using List<T>, and perform operations such as add, display, and search student details.

Assignment 8: Exception Handling Application

Develop a C# console application to demonstrate exception handling by accepting two numbers from the user and performing division. Handle divide-by-zero and format exceptions using try-catch-finally blocks.

Assignment 9: Windows Form – Simple Registration Form

Design a Windows Form application using GUI controls (TextBox, RadioButton, CheckBox, ComboBox, Button) to collect user registration details and display entered values using MessageBox on button click.

Assignment 10: Database Connectivity using ADO.NET

Develop a C# Console/Windows application to connect to SQL Server perform insert, update, delete and select operations

Course Code: CA-315
Course Title: Practical based on JAVA Programming

Course Code: CA-315	Course Category: DSC-19
Course Title: Practical based on JAVA Programming	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To develop fundamental programming skills using core Java concepts. 2. To understand and apply object-oriented programming principles in Java. 3. To implement exception handling and file operations in Java applications. 4. To design and develop simple real-world Java applications.	
Course Outcomes:	Cognitive Level
1. Apply core Java concepts to solve programming problems.	3
2. Design and implement object-oriented Java programs.	4
3. Design simple GUI applications using AWT/Swing..	4
4. Design simple GUI applications with database connectivity	4

Course Content:

Assignments:

1. Write a Java program to display student information.
2. Write a Java program to perform arithmetic operations using switch case.
3. Write a Java program to find the largest element in an array.
4. Write a Java program using command line arguments.
5. Write a Java program to accept user input using Scanner.
6. Write a Java program to demonstrate the use of constructors and methods.
7. Write a Java program to demonstrate method overloading.
8. Write a Java program to demonstrate the concept of inheritance.
9. Write a Java program to demonstrate method overriding.
10. Write a Java Program to Demonstrate the Use of Interface in Java by Implementing Shape Area Calculation.
11. Write a Java Program for String Manipulation Operations.
12. Write a Java Program Using try-catch-finally for Exception Handling.
13. Write a Java Program to Create a Simple Calculator Using AWT/Swing.
14. Write a Java Program to Create a Simple Login Form Using Swing.
15. Write a Java Program to demonstrate database connectivity.
16. Write a Java Program to perform insert, update, delete operations on database.

Course Code: CA-316(A)
Course Title: Web Technologies-I

Course Code: CA-316 (A)	Course Category: DSE-1
Course Title: Web Technology-I (PHP)	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand web architecture and create basic dynamic webpages using PHP. 2. To connect PHP with MySQL and perform basic database operations. 3. To handle user input and implement login authentication using forms and sessions. 4. To apply OOP concepts in PHP and develop simple JSON-based backend services.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Explain web architecture and demonstrate basic PHP scripting.	2
2. Illustrate PHP–MySQL connectivity and interpret database operation results.	2
3. Develop a form-based application with validation and session-based login authentication.	3
4. Create simple object-oriented PHP modules and JSON-based services.	3

Course Content:

Unit 1: Web Fundamentals & PHP Basics

(07 L, 10M)

- Client–Server Architecture.
- HTTP Request and Response.
- Role of Web Server and PHP Interpreter.
- Installing & configuring XAMPP/WAMP.
- Running PHP scripts.
- Server-side scripting and uses of PHP.
- Embedding PHP in HTML.
- PHP syntax structure.
- Output statements (echo, print).
- Variables, Data Types & Constants.
- Operators, Comments.
- Simple conditional statements (if, if-else), Basic loops.

Unit 2: PHP Programming & Database Interaction

(09 L, 15M)

- User-defined functions.
- Arrays (Indexed and Associative) and foreach loop.
- Basic string handling and methods (strlen(), str_word_count(), strtoupper(), strtolower(), trim(), strpos(), substr(), str_replace(), strcmp() etc.).
- Introduction to database and MySQL.
- Using phpMyAdmin.
- Creating database and tables.
- Connecting PHP with MySQL using mysqli.
- Executing SQL queries from PHP.
- CRUD operations: Insert, Select, Update, Delete.
- Displaying database records in HTML table.

Unit 3: Forms, Sessions & Authentication**(08 L, 15M)**

- HTML <form> tag and attributes.
- GET and POST methods.
- \$_GET and \$_POST handling.
- Basic form validation, include and require.
- Cookies and sessions (session lifecycle).
- Login and logout mechanism.
- Password hashing (password_hash, password_verify).
- Basic error handling (die, exit).
- File upload handling.

Unit 4: OOP & Modern PHP Concepts**(06 L, 10M)**

- Class and object in PHP, Constructor.
- Access modifiers (public, private, protected).
- Basic inheritance- Types of Inheritance.
- Basic exception handling (try-catch).
- JSON handling (json_encode, json_decode).
- PHP with JavaScript (AJAX/fetch concept).
- Creating simple JSON-based API response.
- Introduction to MVC architecture.
- Overview of Laravel framework and Composer (conceptual).

Reference Books:

1. **Robin Nixon** — Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5, O'Reilly Media, ISBN-13: **9781491978917**
2. **Jon Duckett** — PHP & MySQL: Server-Side Web Development, Wiley, ISBN-13: 9781119149224.
3. **Luke Welling & Laura Thomson** — PHP and MySQL Web Development (5th Edition), Pearson Education, ISBN-13: 9789332582736
4. **Ivan Bayross & Sharanam Shah** — PHP for Beginners, SPD, ISBN: 9788184040753
5. **Steven Holzner** - PHP: The Complete Reference , McGraw-Hill, ISBN: 978-0070223622
6. **Michael K. Glass et al.** — Beginning PHP, Apache, MySQL Web Development (Wrox, 2004), ISBN: 978-0070223622

Course Code: CA-316(B)
Course Title: Data Analytics-I

Course Code: CA-316 (B)	Course Category: DSE-1
Course Title: Data Analytics –I (R and Tableau)	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: - To provide students with a strong foundation in R programming and Tableau for data analysis and visualization. - To develop student's competency in data manipulation, data visualization, and basic statistical analysis techniques. - To enable student's to apply R programming skills to real-world data analysis problems. - To equip students with practical knowledge of Tableau's features and tools for creating interactive dashboards and visual reports.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Understand the fundamentals of R programming and Tableau, including data types, structures and basic functions for data analysis.	2
2. Apply R programming and Tableau tools to manipulate, clean, and visualize datasets for meaningful insights.	3
3. Analyze data using descriptive statistics, graphs, and interactive dashboards to support data-driven decision-making.	4
4. Analyze the results of data analysis and visualizations to solve real-world problems effectively.	4

Course Content:

UNIT I: Introduction to R Programming

(8 L, 13M)

- Introduction to Data Analytics
 - Introduction to R Programming Language
 - Features of R programming
 - Introduction to R Environment & Workspace
 - Data Types in R: Numeric, Integer, Character, Logical
 - Variables and Operators
 - Basic Input and Output Functions :
print(), cat(), readline(), message(), warning()
 - Vectors : Introduction, Creation, Generating Sequence, Accessing Elements in Vector.
 - List: Introduction, List Creation, Accessing elements.
 - Data Frames: Introduction, Creation, Accessing elements.
 - Importing Data from: CSV / Excel files.

UNIT II: Data Structures & Data Handling in R Programming

(8 L, 12M)

- Conditional Statements: if, if-else
- Loops: for loop, while loop
- Introduction R Packages : dplyr and ggplot2
- Inbuilt functions:
 - Mathematical Function
 - Text Functions
 - Sequence Function
- Sorting and Filtering Data

- Quartiles, Quantiles, Interquartile, Mean, Median
- Introduction to Data Visualization
- Introduction to Box Plot, Pie Chart, Line Chart, Scatter Plot in R

Unit III: Introduction to Data Analytics using Tableau

(7 L, 13M)

- Introduction to Tableau .
- Tableau interface and workspace basics
- Importance of Visual Analytics
- Data types, fields, measures & dimensions
- Connecting Tableau to different data sources (Excel, Text files)
- Introduction to Data interpreter & metadata
- Joins & Relationships
- Data cleaning basics: Filters, Groups, Sets

Unit IV: Basic Visualization Techniques in Tableau

(7 L, 12M)

- Charts & Graphs in Tableau
 - a. Bar, Line, Pie, Tree Map
 - b. Scatterplots, Histograms
- Sorting & Organizing views
- Introduction to Show Me feature
- Basic Principles of Dashboard Design
- Building Dashboards using Dataset
- Layouts, Containers, Objects
- Interactivity: Filters, Actions

References

- **R for Data Science** – Hadley Wickham, Garrett Grolmund & Mine Çetinkaya-Rundel; O'Reilly Media; 2nd Edition (2023); ISBN-13: 9781492097402
- **The Art of R Programming: A Tour of Statistical Software Design** – Norman Matloff; No Starch Press; 1st Edition (2011); ISBN-13: 9781593273842
- **The Book of R: A First Course in Programming & Statistics** – Tilman M. Davies; No Starch Press; 1st Edition (2016); ISBN-13: 9781593276515.
- **R in Action: Data Analysis and Graphics with R** – Robert I. Kabacoff; Manning Publications; (Edition varies); ISBN-13: 9781617291388.
- **Learning Tableau 2020 (or latest edition)** – Joshua N. Milligan; Packt Publishing; (Latest Edition varies); ISBN-13: 9781800200364.
- **Advanced Analytics with R and Tableau** – Jen Stirrup, Ruben Oliva Ramos & Lourdes Bolaños Pérez; Packt Publishing; 1st Edition (2017); ISBN-13: 9781786460110.
- **Mastering Tableau 2023 (or latest)** – Marleen Meier; Packt Publishing; latest edition (2023); ISBN-13: 9781803233765.

Course Code: CA-316(C)
Course Title: Advance Artificial Intelligence Techniques

Course Code: CA-316(C)	Course Category: DSE-1
Course Title: Advanced Machine Learning Techniques	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks

Course Objectives:

1. Understand and apply advanced machine learning techniques including model evaluation, ensemble methods, and feature engineering.
2. Understand and implement deep learning models such as ANN, MLP, CNN, and optimization techniques.
3. Analyze AI applications and evaluate ethical issues including bias, fairness, and responsible AI principles.

Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Apply advanced machine learning techniques.	3
2. Understand deep learning architecture and training methods.	1
3. Analyze NLP and Computer Vision applications.	4
4. Evaluate ethical implications of AI systems.	4

Course Content:

Unit I: Advanced Machine Learning Techniques

(07 L, 10M)

- Review of supervised & unsupervised learning
- Bias-Variance trade off :Underfitting vs Overfitting, Error decomposition
- Model evaluation techniques :K-fold cross-validation, Cross-validation, Confusion matrix
- Ensemble methods:Random Forest, Gradient Boosting
- Feature engineering basics
 - Feature selection methods
 - Feature scaling :Normalization, Standardization
 - Handling missing data

Unit II: Deep Learning Fundamentals

(09 L, 15M)

- Artificial Neural Networks architecture
- Multi-Layer Perceptron(Hidden layers, Forward propagation, Loss functions (MSE, Cross-Entropy)
- Backpropagation algorithm(Chain rule concept, Gradient computation, Weight update mechanism)
- Activation functions (ReLU, Sigmoid, Tanh)
- Optimization algorithms (SGD, Adam)
- Overfitting & Regularization(Dropout, L1 & L2 regularization, Early stopping, Batch normalization)

Unit III: Applications of AI –Text and Image Processing

(09 L, 15M)

- Natural Language Processing:
 - Text preprocessing(Tokenization ,Stop-word removal, Stemming & Lemmatization, N-grams)
 - Sentiment analysis
 - Rule-based Vs. ML-based approaches
 - Feature extraction (TF-IDF)
 - Evaluation metrics
- Image Processing
 - Convolutional Neural Networks (CNN)
 - Basic Image classification concept

Unit IV: Emerging Trends & Ethical AI

(05 L, 10M)

- Reinforcement Learning basics (MDP, Q-learning overview)
- Generative AI introduction
- Large Language Models
 - Transformer architecture overview
 - Pretraining & fine-tuning
 - Prompt engineering basics
 - Applications of LLMs
- AI Ethics:
 - Bias in AI
 - Fairness & transparency
 - Responsible AI principles

References:

1. **S. J. Russell and P. Norvig**, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2020.
2. **I. Goodfellow, Y. Bengio, and A. Courville**, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
3. **C. M. Bishop**, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
4. **A. Géron**, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.
5. **F. Chollet**, *Deep Learning with Python*. Shelter Island, NY, USA: Manning Publications, 2017.
6. **D. Jurafsky and J. H. Martin**, *Speech and Language Processing*, 3rd ed. draft. Stanford, CA, USA: Stanford University, 2023.

Course Code: CA-317(A)
Course Title: Practical based on Web Technology-I

Course Code: CA-317 (A)	Course Category: DSE-2
Course Title: Practical based on Web Technology-I (PHP)	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To familiarize students with the PHP development environment and server configuration. 2. To develop programming skills in PHP for creating dynamic web pages. 3. To enable students to interact with MySQL databases using PHP scripts. 4. To train students to build basic web applications with forms, sessions, and file handling.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. To execute PHP programs on a local server environment	2
2. To develop dynamic webpages using PHP programming constructs.	3
3. To develop dynamic webpages using PHP programming constructs.	3
4. To construct a simple web application with authentication, sessions and file upload features.	3

Assignment List:

1. Install XAMPP server and execute a basic PHP program to display “Hello World” and current date.
2. Write a PHP program using variables and operators to perform simple arithmetic calculations.
3. Implement conditional statements and loops in PHP to check number conditions and generate tables.
4. Create user-defined functions and demonstrate indexed and associative arrays using foreach.
5. Connect PHP with MySQL database and insert student records using URL parameters (without form).
6. Perform CRUD operations by displaying, updating and deleting database records through PHP scripts.
7. Design an HTML form and retrieve user input using \$_GET and \$_POST with basic validation.
8. Develop a login and logout system using sessions for user authentication.
9. Upload a file using PHP and store user information using cookies.
10. Create a PHP class and generate database data in JSON format using json_encode.

Suggested Activities:

Mini Project 1: Online Complaint / Query Submission System

Problem Statement: Design a small web application where users can submit complaints or queries and an admin can view them.

Requirements:

1. Create MySQL database complaint_db
2. Table fields: id, name, email, subject, message, date
3. Create complaint submission form (HTML + PHP)
4. Validate empty fields and email format
5. Store data in database using PHP
6. Display all complaints on admin page
7. Show submission date and time
8. Use session to restrict admin page access (simple login)
9. Provide delete option for complaints

Mini Project 2: Simple Student Result Viewer

Problem Statement:

Create a web application where a student can enter roll number and view his/her result.

Requirements:

- Create MySQL database result_db
- Table fields: roll_no, name, subject1, subject2, subject3, total
- Create search form to enter roll number
- Fetch student data from database using PHP
- Display marks and total in formatted table
- Handle invalid roll number (error message)
- Use PHP functions and string handling
- Provide JSON output option (API page returning result using json_encode)

Course Code: CA-317(B)
Course Title: Practical based on Data Analytics-I

Course Code: CA-317 (B)	Course Category: DSE - 2
Course Title: Practical based on Data Analytics-I	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand R basics and create/manipulate data frames with sorting and filtering. 2. To perform operations on vectors and lists, including creation, sequences, and accessing elements. 3. To create plots using ggplot2 for effective data visualization. 4. To import, combine, and filter data, and create interactive dashboards.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Apply fundamental R programming concepts including data types, data structures, data import techniques and built-in functions for basic data analysis.	3
2. Analyze datasets in R using data manipulation techniques, descriptive statistics, and data visualizations.	3
3. Apply R programming and Tableau tools to clean, organize, and analyze datasets effectively.	3
4. Analyze datasets in R and Tableau to generate insights, interpret results, and create interactive visualizations and dashboards.	4

Course Content:

Assignments of R Programming:

1. Write simple R programs using basic syntax.
2. Create & Manipulate data frames, including sorting and filtering datasets.
3. Import Data from CSV / Excel file and display records in R.
4. Demonstrate Vectors: Create Vectors, Generating Sequences & Accessing Elements.
5. Demonstrate Lists: Create List and Access Elements from Lists.
6. Demonstrate Built-in Functions: i) Mathematical Function ii) Text Function
7. Calculate Central Tendency Measures: Mean and Median.
8. Compute Descriptive Statistics - Quartiles and Quantiles
9. Demonstrate Plot using dataset - i) Line ii) Box iii) Pie iv) Scatter.

Assignments of Tableau:

1. Install Tableau Public and connect to simple datasets such as CSV and Excel files for visualization.
2. Import and combine multiple data sources using join to prepare data for analysis.
3. Apply filters to organize and focus on data sets.
4. Create various charts / graphs including bar, line, pie, scatter etc.
5. Design interactive dashboards by arranging charts, using containers, and implementing interactivity features like filters and actions.

Course Code: CA-317(C)

Course Title: Practical based on Advance Artificial Intelligence Techniques

Course Code: CA-317(C)	Course Category: DSE - 2
Course Title: Practical based on Advance Artificial Intelligence Techniques	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Understand and implement data preprocessing techniques. 2. Develop and evaluate ML/DL models with optimization and regularization. 3. Analyze and apply advanced AI architectures like Transformers in NLP and Computer Vision.	
Course Outcomes:	Cognitive Level
1. Apply data preprocessing and evaluate machine learning models using appropriate metrics and validation techniques.	3
2. Design and optimize ANN models with regularization techniques.	5
3. Explain Transformer architecture and its applications in Large Language Models.	2

Course Content:

Prerequisite Knowledge

- **Python Programming:**
Basic knowledge of Python, data structures, functions, and working with NumPy and Pandas.
- **Mathematics & Statistics:**
Understanding of matrices, derivatives (basic calculus), probability, mean, variance, and standard deviation.
 1. Write a program to perform data preprocessing by handling missing values in a dataset.
 2. Write a program to encode categorical variables using Label Encoding and One-Hot Encoding.
 3. Write a program to apply feature scaling using Normalization and Standardization techniques.
 4. Write a program to evaluate a classification model using Confusion Matrix, Accuracy, Precision, Recall, and F1-score.
 5. Write a program to implement K-Fold Cross-Validation and compare results with train-test split.
 6. Write a program to implement the ensemble method – Random Forest – and analyze feature importance.
 7. Write a program to implement a simple Artificial Neural Network (ANN) for binary classification.
 8. Write a program to implement a Multi-Layer Perceptron (MLP) with at least one hidden layer.
 9. Write a program to implement the Backpropagation algorithm and update weights using gradient descent.
 10. Write a program to train a neural network using SGD optimizer and compare performance with Adam optimizer.
 11. Write a program to apply L1 and L2 regularization in a neural network and compare results.
 12. Write a program to implement Dropout regularization and analyze its impact on model performance.
 13. Write a program to build a Convolutional Neural Network (CNN) for image classification.

Required Tools

- Jupyter Notebook/Google Colab/VS Code / PyCharm

Course Code: CA-318(A)
Course Title: Financial Management

Course Code: CA-318 (A)	Course Category: Minor-6
Course Title: Financial Management	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Understand core concepts of financial management, objectives and decisions. 2. Comprehend the financial statements using comparative tools and methods 3. Know the key ratios for liquidity, profitability and efficiency assessment 4. Explain working capital dynamics and determinants in IT/service contexts	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Define financial management scope and distinguish it from accounting	1
2. Identify components of financial statements and understand analysis methods	2
3. Classify and explain major ratios to know about the business financial health.	3
4. Describe working capital needs using operating cycle in service industries	3

Course Content:

Unit 1: Introduction to Financial Management (Theory Only)

(07 L, 08M)

- Meaning and Definition of Financial Management
- Objectives of Financial Management- Primary and Secondary Objectives.
- Functions of Financial Management
- Importance of Financial Management in Business
- Financial Decisions – Investment decisions, Financing decisions and Dividend decisions

Unit 2: Financial Statement & their Analysis (Theory Only)

(09 L, 15M)

- Meaning, Objectives and Importance of Financial Statements.
- Types of Financial Statements- Balance Sheet, Income Statement (Profit & Loss A/c), Cash Flow Statement and their Components.
- Meaning & Objectives of the Financial Statement Analysis
- Methods of Analysing Financial Statements- Comparative Statements, Common-size statements and Trend analysis.

Unit 3: Ratio Analysis (Theory Only)

(08 L, 12M)

- Meaning, Objectives and Importance of the Ratio Analysis
- Classification of Ratios – Liquidity, Profitability, Solvency and Efficiency
- **Liquidity Ratios** – Current ratio, Quick ratio and Cash ratio
- **Profitability Ratios** – Operating profit ratio, Gross profit ratio and Net profit ratio

Unit 4: Working Capital Management (Theory Only)

(06 L, 10M)

- Meaning and Importance of the Working Capital
- Types of Working Capital – Gross vs. Net Working Capital; Permanent vs. Temporary Working Capital.
- Working Capital Cycle – Operating Cycle concept with special reference to IT and Service Industries.
- Determinants of Working Capital Requirements

Reference Books:

- Kishore, R. M., & Arora, P. S. (2023). Taxmann's financial management: Theory, problems & cases (9th ed.). Taxmann Publications. ISBN: 9789357780841
- Chandra, P. (2023). Financial management: Theory and practice (10th ed.). McGraw Hill Education. ISBN:

9789355327560.

- Brigham, E. F., & Houston, J. F. (2022). Fundamentals of financial management (16th ed.). Cengage Learning. ISBN: 9780357033831
- Pandey, I. M. (2021). Financial management (11th ed.). Vikas Publishing House. ISBN: 9789325991848.

Course Code: CA-318(B)
Course Title: Linear Algebra-I

Course Code: CA-318(B)	Course Category: Minor-06
Course Title: Linear Algebra-I	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Apply matrix methods to solve systems of linear equations. 2. Apply vector operations and geometry in R^n 3. Use matrices and linear transformations to solve applied problems. 4. Analyze determinants, eigenvalues, and vector space concepts.	
Course Outcomes: After completion of the course, students will be able to:	Cognitive Level
1. Solve and interpret systems of linear equations using matrix techniques.	3
2. Apply vector algebra to analyze lines, planes, and projections.	3
3. Analyze matrix operations and linear transformations.	4
4. Compute and analyze determinants, eigenvalues, eigenvectors, and subspaces.	4

Course Content:

UNIT 1 – Systems of Linear Equations **(7 L, 12 M)**

- Geometric view of linear equations (Lines & Planes)
- Algebraic representation (Matrix form $AX = B$)
- Elementary row operations
- Gaussian elimination
- Gauss–Jordan elimination
- Homogeneous & non-homogeneous systems
- Applications (Balancing equations, Network problems – basic idea)

UNIT 2 – Vectors in R^n and Geometry **(8 L, 13 M)**

- Vectors and coordinate systems
- Vector addition & scalar multiplication
- Linear combinations
- Dot product and its properties
- Length (Norm) and distance
- Cauchy–Schwarz inequality (statement & meaning)
- Orthogonality and projections
- Lines and planes in R^2 & R^3

UNIT 3 – Matrices and Transformations **(8 L, 12M)**

- Matrix operations
- Matrix multiplication and properties
- Matrix inverse and solving systems
- Elementary matrices
- Transpose of a matrix
- Linear transformations
- Matrix of a linear transformation
- Geometric interpretation (Scaling, Rotation, Reflection)

UNIT 4 – Determinants, Eigenvalues & Vector Spaces

(7 L, 13 M)

- Determinants (2×2 , 3×3)
- Properties of determinants
- Eigenvalues and eigenvectors (basic computation)
- Diagonalization (basic idea)
- Vector spaces and subspaces
- Basis and dimension
- Column space, Row space, Null space

Reference Book:

- **P. Selinger**, *Matrix Theory and Linear Algebra*, open textbook, Dalhousie University, Halifax, NS, Canada.
- <https://www.mathstat.dal.ca/~selinger/linear-algebra/downloads/LinearAlgebra.pdf>.
- **Linear Algebra and Its Applications** – David C. Lay, Steven R. Lay & Judi J. McDonald; Pearson Education; 5th Edition (2015); ISBN-13: 9780321982384
- **Introduction to Linear Algebra** – Gilbert Strang; Wellesley-Cambridge Press; 5th Edition (2016); ISBN-13: 9780980232776
- **Elementary Linear Algebra** – Howard Anton & Chris Rorres; John Wiley & Sons; 11th Edition (2013); ISBN-13: 9781118677452
- **Linear Algebra Done Right** – Sheldon Axler; Springer; 3rd Edition (2015); ISBN-13: 9783319110790
- **Applied Linear Algebra and Matrix Analysis** – Thomas S. Shores; Springer; 2nd Edition (2018); ISBN-13: 9783319747545

Course Code: CA-319(A)
Course Title: Practical based on Financial Management

Course Code: CA-319(A)	Course Category: Minor -07
Course Title: Practical based on Financial Management	Type: Practical
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To develop the ability to prepare and interpret comparative and common-size financial statements. 2. To enable students to analyze financial performance using trend analysis techniques. 3. To equip students with skills to evaluate liquidity and profitability through ratio analysis. 4. To understand and assess working capital requirements for efficient business operations.	
Course Outcomes: After completion of the course, students will be able to:	Cognitive Level
1. Prepare and analyze comparative and common-size financial statements.	3
2. Compute and interpret trend percentages for financial performance evaluation.	4
3. Analyze liquidity and profitability using appropriate financial ratios.	4
4. Compute working capital and operating cycle of the business.	4

Course Content:

- 1. Practical problems on Comparative Financial Statements:** Students will prepare and analyze comparative financial statements (Balance Sheet and Income Statement) for multiple periods.
- 2. Practical problems on Common-Size Financial Statements:** Students will convert financial statements into common-size format by expressing each item as a percentage of a base figure.
- 3. Practical problems on Trend Analysis:** Students will compute trend percentages using a base year to evaluate long-term movements in financial data.
- 4. Practical problems on Liquidity Ratios:** Students will calculate and interpret liquidity ratios such as current ratio, quick ratio and cash ratio.
- 5. Practical problems on Profitability Ratios:** Students will analyze profitability through ratios like operation ratio, gross profit ratio and net profit ratio.
- 6. Practical problems on Working Capital Requirements:** Students will calculate operating cycles, assess components of working capital and determine adequate levels required for smooth business operations.

Note: The faculty members are instructed to teach at least two practical problems for each topic. Problems should preferably be based on company's financial statements to ensure practical exposure.

Reference Books:

- Kishore, R. M., & Arora, P. S. (2023). Taxmann's financial management: Theory, problems & cases (9th ed.). Taxmann Publications. ISBN: 9789357780841
- Chandra, P. (2023). Financial management: Theory and practice (10th ed.). McGraw Hill Education. ISBN: 9789355327560.
- Brigham, E. F., & Houston, J. F. (2022). Fundamentals of financial management (16th ed.). Cengage Learning. ISBN: 9780357033831
- Pandey, I. M. (2021). Financial management (11th ed.). Vikas Publishing House. ISBN: 9789325991848.

Assessment Criteria:**Marking Scheme – College Assessment (CA): Total 20 Marks**

- Solve Any Two out of three practical problems.
- Written Solution for the practical problems: 10 Marks (05 Marks each practical)
- Journal File: 10 Marks

Marking Scheme - University Assessment (UA): Total 30 Marks

- Solve Any Two out of three practical problems.
- Written Solution for the practical problems: 20 Marks (10 Marks each practical)
- Journal File: 10 Marks

Course Code: CA-319(B)
Course Title: Practical on Linear Algebra -I

Course Code: CA-319(B)	Course Category: Minor-07
Course Title: Practical Linear Algebra-I	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Apply programming techniques to solve linear algebra problems. 2. Implement matrix and vector operations using computational methods. 3. Analyze linear transformations and eigenvalue problems through coding. 4. Develop computational solutions for applied linear algebra concepts.	
Course Outcomes: After completion of the course, students will be able to:	Cognitive Level
1. Implement Gaussian elimination and Gauss–Jordan methods to solve linear systems programmatically.	3
2. Develop programs to perform matrix and vector operations including determinant, inverse, and projections.	3
3. Apply coding techniques to verify span, orthogonality, and linear transformations.	3
4. Compute and analyze eigenvalues and eigenvectors using computational tools.	4

Prerequisites for Linear Algebra

Students should have basic knowledge of:

- Mathematics: Basic algebra, coordinate geometry, matrices & determinants
- Programming: Variables, loops, conditions, functions
- Python Basics: Syntax, indentation, lists/arrays, importing libraries
- Basic Math: Vectors, matrices, linear equations, matrix multiplication

Software Tools Required

- Programming Environment: Python 3.x (≥ 3.9 recommended)
- Libraries: NumPy, Matplotlib, Pandas, Scikit-learn, Pillow (PIL)

Practical Assignments:

1. **Write a program for** solving linear systems using Gaussian elimination.
2. **Write a program for** implementing Gauss–Jordan row reduction.
3. **Write a program for** performing matrix operations (add, multiply, transpose, determinant, inverse).
4. **Write a program for** computing vector operations (dot product, magnitude, angle, plotting).
5. **Write a program for** checking whether a vector belongs to span of given vectors.
6. **Write a program for** checking orthogonality and computing vector projection.
7. **Write a program for** applying linear transformations (scaling, rotation, reflection).
8. **Write a program for** computing eigenvalues and eigenvectors.

Course Code: CA-320

Course Title: Software Development using recent tools and techniques

Course Code: CA320	Course Category: VSC-4
Course Title: Software Development using Recent tools and techniques	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand the need for software development tools. 2. To learn version control system and collaboration tool GitHub 3. To understand Project Management Using Jira . 4. To Study Cloud-Native Development	
Course Outcomes: After completion of the course, students will be able to:	Cognitive Level
1. Understand software development tools Agile and DevOps.	2
2. Explain the importance of UI/UX design in software development.	2
3. Describe the introduction and features of Figma	2
4. Explain the basic concepts of cloud platforms and Cloud-Native Development	2

Course Content:

Unit 1 Introduction to Modern Software Development

(07 L, 10M)

- The evolving role of software
- Layered technology and process framework
- Need of software development tools: Improve efficiency, Enhance quality, Support collaboration, Reduce risk, Enable faster delivery
- Agile : Agile Principles, Plan-driven and Agile development, Extreme programming, Scrum, A Tool Set for the Agile Process.
- DevOps : Introduction, DevOps Lifecycle

Unit 2: Version Control and Collaboration Tools

(09 L, 15M)

- Version Control Systems (VCS):
 - Introduction,
 - Need for Version Control System
 - Types of Version Control Systems-
- Local Version Control,
- Centralized Version Control(CVCS),
- Distributed Version Control (DVCS)
- Introduction to Git : Basic Git Commands (init, add, commit, push, pull)
- Introduction to GitHub :
 - Concept of GitHub
 - Features of GitHub
 - Working with GitHub-Creating GitHub Account, Creating a Repository,Cloning
 - Repository, Pushing Changes, Pulling Changes

Unit 3: Recent Development Technology and tools

(7 L, 15M)

- Introduction of Software development technology.-
 - Front-End Technologies.
 - Back-End Technologies.
 - Database Technologies.
- UI/UX Design
 - Figma - introduction, Features of Figma
- Testing & Quality Assurance Tool
 - Postman - Introduction, Features of Postman, Advantages of Postman, Role of Postman in Testing
- Project Management Tool
 - Jira -Introduction, Features, Types of Projects in Jira

Unit 4: Cloud & Deployment Basics

(7 L, 10M)

- Introduction to Cloud Computing
- Cloud platforms –
 - Amazon Web Services.
 - Microsoft Azure.
 - Google Cloud Platform
- Introduction to Cloud-Native Development
 - Microservices,
 - Containers,
 - Serverless,
 - Kubernetes
 - Docker

Reference Books

1. GitHub Essentials – by, Achilleas Pipinellis – 1st Edition – ISBN: 978-1783553754

2. API Testing and Development with Postman, by Dave Westerveld -1st Edition

ISBN: 978-1801817188

3.Mastering Cloud Computing – Rajkumar Buyya, Christian Vecchiola & S. Thamarai Selvi.

McGraw Hill Education

Course Code: CA-310
Course Title: Field Project

Course Code: CA-310	Course Category: (FP-2)
Course Title: Field Project	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To apply research methodology for identifying and analysing real-time problems. 2. To analyze system requirements and design appropriate software architecture. 3. To develop and implement a functional software solution 4. To evaluate system performance, usability, and feasibility and analysis techniques.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level [Bloom's
1. Apply research-based problem-solving approaches to identify IT challenges.	3
2. Analyze functional and non-functional requirements to design structured system models and database schemas.	4
3. Develop and implement a complete software system using programming languages and database technologies.	3
4. Analyze system performance and compare existing and proposed solutions	4

GUIDELINES FOR FIELD PROJECT FORMULATION

- Each student shall have to submit a Field Project during 5th Semester.
- In the 5th semester examination student are required to carry out a Field Project individually or by group of four students.
- The Field Project should be based on **requirement gathering and system analysis for the development of a computer-based application.**
- Students may select a Field project from any one of the following categories:

1. Study of an Existing System

Students may visit an organization,

- Schools, Colleges, Shops, Medical Shops, Various Government Offices, Hospitals, Banks, Small and Medium Industries etc. to study how an existing system works.
- Students should interact with the users of the system and collect information through surveys, interviews, or observation to understand the current working process and identify the problems or limitations faced by users.
- Based on the collected data and analysis, students should propose and develop a suitable software application as an improved solution for the existing system.

2. Problem-Based System Development

- Students may identify a general problem in society or in an organization where a proper system is not available.
- Students should conduct a survey, understand the requirements, and then develop an application to solve the problem.

Steps to Conduct the Field Project

Step 1: Problem Identification and Feasibility Study

Students must identify a **real-world problem** or select an **existing system requiring improvement**.

Activities include:

- Understanding the existing workflow
- Identifying system limitations
- Determining feasibility of a computerized solution

Step 2: Requirement Analysis

Students must collect and analyse **user requirements**.

Methods of requirement gathering may include:

- User Interviews
- Observation
- Questionnaires
- Surveys

Activities include:

- Identifying functional requirements
- Identifying non-functional requirements
- Defining system scope and objectives

Step 3: System Design

Students must design the proposed software system using standard **system modeling techniques**.

Design activities include:

- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- Use Case
- Database Schema Design
- User Interface Design

Step 4: Application Development

Students must develop a simple software application based on analysed the collected requirements.

Step 5: Testing and Validation

- Manual Testing

Step 6: Field Project Report

Students must prepare a **complete Field project report** documenting all phases of software development.

Documentation includes:

- Problem analysis
- Requirement specification
- System design
- Implementation details
- Conclusion and future scope

Project Report Format

The project report should include the following chapters:

Chapter 1: Introduction

- Background of the project
- Objectives of the project

Chapter 2: Problem Identification and Feasibility Study

- Description of the current system
- Limitations or problems in the system

Chapter 3: Requirement Analysis

- Method of data collection
- Field visit details
- Interview details
- Questionnaire / survey method

- Observation details
- Summary of collected data

Chapter 4: System Design

- Flowcharts
- ERD, DFD
- Database design

Chapter 5: Implementation

- Technologies used
- Application screenshots
- Description of modules

Chapter 6: Conclusion and Future Scope

General Guidelines

- The Field Project topic must be selected in consultation with the Project Guide .
- The Field Project must be based on requirement analysis and primary data acquisition obtained through stakeholder interaction and system observation
- Students must perform system study and requirement gathering, by interacting with users or stakeholders of the selected organization or domain.
- User requirements should be gathered through appropriate data collection techniques such as interviews, observation, questionnaires, or surveys.
- Students must analyze the collected requirements and develop a computer-based application system following appropriate Software Development Life Cycle (SDLC) phases.
- Students must prepare a comprehensive Project Report according to the prescribed format, including system analysis, system design, implementation, and testing details.
- Each project group must submit two printed copies of the final Project Report .
- **Study of an Existing System:** Students must obtain an Organization Certificate from the organization or institution where the system study or requirement analysis was conducted.
- **Problem-Based System Development:** The college/institute shall issue a certificate duly signed by the mentor.
- The Field Project will be evaluated for 50 marks, out of which 20 marks will be allotted for internal assessment by the guide and 30 marks for external evaluation based on the vivo-voce

Assessment:

Internal Assessment (20 Marks)

- Continuous evaluation by the project guide.

External Assessment Criteria

Component	Marks
Problem Analysis & Requirement Study	6
System Design	6
Application Development	6
Field Project Report	6
Viva / Demonstration	6

SEMISTER VI

Course Code: CA-321
Course Title: Cloud Computing

Course Code: CA-321	Course Category: DSC-20
Course Title: Cloud Computing	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand fundamental concepts, service models, and deployment models of cloud computing. 2. To explain cloud architecture, virtualization technologies, and networking concepts. 3. To apply cloud security mechanisms and risk management strategies. 4. To analyze cloud applications and emerging technologies in modern computing environments.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Describe cloud computing concepts, service models, and deployment models.	2
2. Analyze cloud architecture, virtualization techniques, and networking components.	4
3. Implement security mechanisms and risk mitigation strategies in cloud environments.	3
4. Analyze cloud applications and emerging technologies such as AI/ML, IoT, and containerization.	4

Course Content:

Unit 1: Fundamentals of Cloud Computing

(09 L, 10 M)

- Introduction to Distributed and Cloud Computing.
- Roots and Evolution of Cloud Computing.
- Characteristics and Features of Cloud
- Types of Cloud: Public, Private, Hybrid.
- Cloud Service Models: IaaS, PaaS, SaaS.
- Benefits and Limitations of Cloud Computing.
- Cloud Service Providers: Amazon Web Services, Microsoft Azure, Google Cloud Platform.

Unit 2: Cloud Architecture and Virtualization

(08L, 15 M)

- Cloud Computing Architecture.
- Introduction to Virtualization Technologies and its types.
- Virtualization Technologies: Virtual Machines, Hypervisors, Provisioning and Migration Techniques.
- Load Balancing in Cloud Computing
- Networking in Cloud: Virtual Private Cloud (VPC), Subnets.
- Cloud Computing vs. Network Computing.

Unit 3: Cloud Security and Risk Management

(08 L, 10 M)

- Cloud Security Architecture and Shared Responsibility Model.
- Identity and Access Management (IAM), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC).
- Data Security, Encryption Mechanisms.
- Vendor Lock-in and Multi-Cloud Risk Management.
- Data Loss Risks, Backup, and Data Recovery Strategies.
- Security Threats: Malware, Ransomware, DDoS, Phishing, and Insider Attacks.

Unit 4: Cloud Applications and Emerging Trends

(05 L, 15 M)

- Cloud Applications: Business, Industry, Education, Healthcare, and Smart Cities.
- Basics of AI/ML, Big Data, IoT, and Edge Computing in Cloud.
- Containerization and Microservices Architecture.
- Emerging Trends Introduction: 5G Integration, Green Cloud, Block chain-as-a-Services.

References:

1. **Sosinsky B.**, “Cloud Computing Bible”, Wiley India ISBN 13: 9788126529803.
2. **Buyya R., Broberg J., Goscinski A.**, “Cloud Computing: Principles and Paradigm”, John Wiley & Sons ISBN NO: 81-7758-575-4.
3. **Velte T., Velte A., Elsenpeter R.**, “Cloud Computing – A practical Approach”, Tata McGrawHill. ISBN 13: 9780070683518.
4. **Barrie Sosinsky**: "Cloud Computing Bible", Wiley-India, 2010 .
5. **Nikos Antonopoulos, Lee Gillam**: "Cloud Computing: Principles, Systems and Applications", Springer, 2012.
7. **Ronald L. Krutz, Russell Dean Vines**: "Cloud Security: A Comprehensive Guide to Secure Cloud Computing", Wiley-India, 2010.
8. **Tim Mather, Subra Kumara swamy, Shahed Latif**, **Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance**, O'Reilly Media, 2009.
9. **Anthony T. Velte, Toby J. Velte & Robert Elsenpeter**-Cloud Computing: A Practical Approach – Pearson Education.
10. **Rajkumar Buyya, Christian Vecchiola & S. Thamarai Selvi**-Mastering Cloud Computing . McGraw Hill Education.
11. **Thomas Erl, Ricardo Puttini & Zaigham Mahmood**-Cloud Computing: Concepts, Technology & Architecture – Pearson Education.

Course Code: CA-322
Course Title: Recent Scripting Languages

Course Code: CA-322	Course Category: DSC-21
Course Title: Recent Scripting Languages	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To introduce students to React and develop basic interactive user interfaces using components and forms. 2. To enable students to create server-side programs using Node.js runtime and core modules. 3. To train students to design RESTful web services and handle HTTP requests using Express.js. 4. To familiarize students with NoSQL databases and perform database operations using MongoDB with Node.js.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Explain React concepts and demonstrate basic component structure and form.	2
2. Develop basic server-side applications using Node.js modules and HTTP server.	2
3. Implement REST API endpoints and process HTTP requests using Express.js.	3
4. Integrate MongoDB with Node.js and perform CRUD database operations.	3

Course Content:

Unit 1: React Fundamentals

(07 L, 10 M)

- Need of frontend frameworks.
- SPA (Single Page Application) concept.
- Introduction to React.
- Setting up React project
- Introduction of Virtual DOM.
- JSX syntax.
- Components.
- State.
- Props, Props Validation with datatypes.
- Routes.
- React Forms: Controlled Components, Input Elements.
- Event handling.

Unit 2: Node.js Fundamentals

(8 L, 15 M)

- JavaScript runtime outside browser.
- Introduction to Node.js.
- Installing Node.js & npm.
- Node REPL.
- Modules .
- Export and require.
- File handling using fs.
- Creating a basic HTTP server.
- Handling request and response.
- Express Params and Query String.

Unit 3: Express.js & REST API Development

(07 L,10 M)

- Introduction to Express.js.
- Installing packages using npm.
- Routing.
- GET and POST requests.
- Handling JSON data.
- Middleware concept (basic).
- REST API design.
- Creating a simple CRUD API.
- Testing API using Postman.

Unit 4: MongoDB with Node.js

(08 L, 15 M)

- Introduction to NoSQL databases.
- Difference between SQL and NoSQL databases.
- MongoDB concepts: database, collection, document.
- Installation of MongoDB and local configuration.
- Basic MongoDB operations: insert, find, update, delete.
- Introduction to Mongoose.
- Connecting Node.js with MongoDB.
- Performing CRUD operations using MongoDB.
- Storing and retrieving data from a database.

Reference Books:

1. **Stoyan Stefanov** — React: Up & Running (2nd Edition), O'Reilly Media, ISBN-13: 978-1492051466
2. **Alex Banks & Eve Porcello** — Learning React (2nd Edition), O'Reilly Media, ISBN-13: 978-1492051725
3. **Azat Mardan** — Practical Node.js: Building Real-World Scalable Web Apps, Apress, ISBN-13: 978-1484200384
4. **Vasan Subramanian** — Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Apress, ISBN-13: 978-1484239773.
5. **Greg Lim** — Beginning MERN Stack: Build and Deploy a Full Stack MongoDB, Express, React, Node.js App, Apress, ISBN-13: 978-1484257890
6. **Dhruti Shah** — Node.js Guidebook, BPB Publications, ISBN-13: 978-9387284432

Course Code: CA-323
Course Title: Mobile Application Development

Course Code: CA-323	Course Category: DSC-22
Course Title: Mobile Application Development	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand the fundamentals of mobile computing, cellular networks, and Android operating system. 2. To understand and configure the Android development environment and explain the Android architecture. 3. To design and develop Android applications using layouts, widgets, and resources. 4. To understand activity lifecycle, intents, menus, and fragments in Android. 5. To Implement and manage data storage in Android applications using SQLite.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Explain the concepts of mobile computing, cellular networks, and Android OS architecture.	2
2. Set up Android development environment and create a basic Android project.	3
3. Design Android application layouts using Linear, Relative, Constraint, Frame, and Table Layouts.	3
4. Implement UI widgets, menus, and fragments to enhance user interaction in Android apps.	3
5. Develop Android applications with working database functionality using SQLite for data storage and retrieval.	3
6. Analyze and troubleshoot Android application behavior using activity lifecycle, intents, and debugging tools.	4

Course Content:

Unit 1: Mobile Computing and Android Overview

(08 L, 12 M)

- Mobile Computing: Introduction, Applications, Limitations, Architecture
- Characteristics of Mobile Communication
- Mobile Application Types: Native, Hybrid, Web
- Cellular Overview: Cellular networks, Location Management, Handoffs, Noise
- GSM, CDMA, FDMA, TDMA
- Android OS Overview and History
- Android Features, Google Play Store
- Android Architecture, Dalvik Virtual Machine, .apk file
- Android Devices, Versions, API Levels
- Setting up Android Development Environment and SDK

Unit 2: Android Application Structure and Layouts

(07L, 13 M)

- Android Project Structure: Manifest.xml, .java files
- Design Criteria: Hardware Considerations, Design Demands
- Layouts: Linear Layout, Relative Layout, Constraint Layout, Frame Layout, Table Layout
- Drawable Resources.
- Intents, Activity, Activity Lifecycle
- First Sample Android Application

Unit 3: Android UI Design & User Interaction**(08 L, 12 M)**

- Views and ViewGroups
- Layouts: LinearLayout, RelativeLayout, ConstraintLayout
- UI Widgets:-TextView, EditText, Button, CheckBox, RadioButton, ToggleButton, ImageView
- ListView / RecyclerView, ScrollView (Vertical & Horizontal), Event Handling & Listeners
- Toast & Dialog, Menus: Options Menu, Context Menu, Popup Menu

Unit 4: Android Database – SQLite**(07 L, 13 M)**

- Introduction to SQLite
- SQLiteOpenHelper and Database Creation
- Opening and Closing a Database
- Working with Cursors
- Insert, Update, Delete Operations

Reference Books:

1. **Android Application Development** (Black Book),Pradeep Kothari Publication: Dreamtech Press (India), ISBN: 978-9351194094
2. **Android Application Development**, Wei-Meng Lee,Beginning, Publication: Wiley, ISBN: 978-1118199540.
3. **Android Application Development**, Barry,Burd, Wiley India Pvt.Ltd., ISBN: 978-93-5424-5787
4. **Android**, Dixit,P.K., VIKAS, PUBLISHING HOUSE, ISBN: 978-93259-7788-4
5. **Beginning Android Programming with Android Studio**, Dimarzio, J. F., Wrox Wiley India Pvt. Ltd., ISBN: 978-81-265-6558-0

Course Code: CA-324
Course Title: Practical Based on Scripting Languages

Course Code: CA-324	Course Category: DSC-23
Course Title: Practical Based on Scripting Languages	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To set up the Node.js development environment and execute basic scripts. 2. To develop server-side programs using Node.js modules and file handling. 3. To build RESTful APIs using the Express.js framework. 4. To integrate MongoDB with Node.js for data storage and retrieval.	
Course Outcomes: After completion of the course, the student will be able	Cognitive Level
1. To run and test Node.js programs on a local system.	2
2. To implement server and file operations using Node.js modules	3
3. To create and test REST APIs using Express.js.	3
4. To perform CRUD operations by connecting Node.js with MongoDB.	3

Assignment List:

1. Install Node.js and create a React project, then develop a simple component to display static information on a webpage.
2. Create a React component that accepts props and updates displayed data using state and event handling (e.g., counter or toggle message).
3. Design a React form using controlled components to accept user input and display submitted data on the same page.
4. Create a Node.js program demonstrating module creation and usage, and perform file read/write operations using the `fs` module.
5. Develop a basic HTTP server in Node.js that responds differently for multiple URLs and handles query parameters.
6. Install Express.js and implement routing to serve different responses for multiple routes using GET requests.
7. Create REST API endpoints to accept POST requests and return JSON responses, and test them using Postman.
8. Install and configure MongoDB locally, create a database and collection, and perform insert and find operations using Node.js.
9. Integrate MongoDB with Express using Mongoose and implement complete CRUD operations through API endpoints.
10. Connect the React application with the Express API to fetch and display database records and add new data from the React form.

Suggested Activities:

Mini Project 1: Student Record Management System

Problem Statement: Develop a web application where users can manage student records online.

Requirements:

- Create React interface to enter student details (name, course, city)
- Display student list in a table
- Create Node.js + Express backend server
- Implement REST API:
 - Add student
 - View all student
 - Update student
 - Delete student
- Store data in MongoDB collection students
- Fetch and display data in React using API calls
- Validate empty input fields

Mini Project 2: To-Do / Task Manager Application

Problem Statement: Create a simple task management application to maintain daily tasks.

Requirements:

- React page to add new task (title, description)
- Show all tasks in list format
- Mark task as completed
- Delete task option
- Node.js + Express REST API endpoints
- MongoDB collection tasks
- Store and retrieve task data from database
- Use fetch/axios in React to call APIs

Course Code: CA-325**Course Title: Practical Based on Mobile Application Development**

Course Code: CA-325	Course Category: DSC-24
Course Title: Practical based on mobile application Development	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To Explain the process of installing and configuring the Android development environment 2. To develop user interfaces using standard Android widgets. 3. To develop the ability to handle events and user interactions in mobile applications. 4. To Apply input validation techniques and provide appropriate user feedback in Android applications. 5. To Design and integrate navigation, intents, menus, and progress indicators in Android applications.	
Course Outcomes: After completion of the course, the student will be able	Cognitive Level
1. To Configure the Android development environment including JDK, SDK, IDE, and virtual devices for application development.	2
2. To Develop Android applications using standard UI components such as TextView, EditText, Button, and ScrollView.	3
3. To Implement input validation and provide appropriate user feedback using Toast messages and event handling mechanisms.	3
4. To Develop multi-screen Android applications using intents and navigation techniques.	3
5. To Design and construct interactive Android applications such as login systems, calculators, menus, and progress bars.	4

Course Content:**Assignments**

1. Installation and setup of java development kit (JDK), setup android SDK, setup eclipse IDE, setup android development tools (ADT) plugins, create android virtual device.
2. Create "Hello World" application. That will display "Hello World" in the middle of the screen using TextView Widget in the red color.
3. Create Registration page to demonstration of Basic widgets available in android.
4. Create sample application with login module.(Check username and password) On successful login, Change TextView "Login Successful". And on failing login, alert user using Toast "Login fail".
5. Create an application for demonstration of Scroll view in android.
6. Create login application where you will have to validate username and passwords till the username and password is not validated, login button should remain disabled.
7. Create an application for calculator.
8. Create an application to display a list of items using List View.
9. Demonstrate use of scroll view(Vertical and Horizontal).
10. Demonstrate use of intent in android.
11. Create application to demonstrate menu option.
12. Create application to demonstrate progress bar.

Course Code: CA-326 (A)
Course Title: Web Technologies-II (ASP.Net)

Course Code: CA-365 (A)	Course Category: DSE - 3
Course Title: Web Technologies-II (ASP.Net)	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand the fundamentals of web application development using ASP.NET Core. 2. To learn MVC architecture and request–response processing in web applications. 3. To implement database connectivity using Entity Framework Core with Microsoft SQL Server. 4. To develop RESTful Web APIs and understand dependency injection, authentication, and authorization concepts. 5. To prepare students for industry-oriented ASP.NET Core application development.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Understand ASP.NET Core framework, web application architecture, and MVC structure.	2
2. Develop ASP.NET Core MVC applications using controllers, views, routing, and validation techniques.	3
3. Apply Entity Framework Core and LINQ to perform CRUD operations using Microsoft SQL Server.	3
4. Design basic RESTful Web APIs and implement dependency injection and authentication mechanisms.	4

Course Content:

Unit 1: Introduction to ASP.NET and Web Architecture

(07 L, 10 M)

- Introduction to ASP.NET Core and its features
- Overview of Web Application Development using ASP.NET Core: Client–server architecture, HTTP protocol basics, and web request is handled using request–response flow.
- MVC Architecture, Project Structure in ASP.NET Core MVC: Controllers folder, Models folder, Views folder, wwwroot, appsettings.json.
- Routing in ASP.NET Core :Default routing and basic attribute routing concept.
- Introduction to Razor View Engine :Razor syntax basics, use of @Model, @ViewBag.
- Layout Pages and Partial Views :Reuse of common UI components such as header and footer.

Unit 2: ASP.NET Core MVC Fundamentals

(09 L, 15 M)

- Creating Models in ASP.NET Core MVC :Simple model classes with properties.
- Data Annotations for Validation :Required, StringLength, Range – basic usage.
- Controllers and Action Methods :Create controller, define action methods, return View().
- Strongly Typed Views :Model data is passed from controller to view.
- Passing Data from Controller to View :ViewData, ViewBag, TempData – purpose and comparison.
- Views and Razor Syntax :Basic Razor syntax for displaying dynamic data.
- Form Handling in MVC :HTML form creation, form submission, basic model binding.
- Validation Concepts in MVC : Overview Client-side and server-side validation.
- Partial Views and Simple MVC Application Flow .

Unit 3: Database Connectivity using Entity Framework Core with Microsoft SQL Server (07 L, 15 M)

- Introduction to Relational Databases and Microsoft SQL Server.
- Overview of Entity Framework Core – ORM concept and advantages.
- Code First Approach – Model-to-database workflow.
- DbContext and DbSet – Database operations.
- Migrations – Add-Migration and Update-Database.
- CRUD Operations using EF Core with SQL Server.
- One-to-Many Relationships.
- LINQ Queries with EF Core – Where, Select.

Unit 4: Advanced ASP.NET Concepts and Industry Practices (07 L, 10 M)

- Introduction to Web API and difference between MVC and Web API.
- RESTful Service Concepts and HTTP Methods :GET, POST, PUT, DELETE.
- Creating Basic Web API Controller.
- Dependency Injection in ASP.NET Core.
- Authentication and Authorization Overview.
- Error Handling and HTTP Status Codes.
- Overview API Testing using Postman.

Reference Books:

1. **Adam Freeman** — Pro ASP.NET Core MVC (8th Edition), Apress Publications, ISBN-13: 9781484294390
2. **Jon Galloway, Bradley Wilson, K. Scott Allen & David Matson** — Professional ASP.NET Core MVC, Wrox Publications, ISBN-13: 9781119393245
3. **Dino Esposito** — Programming ASP.NET Core, Microsoft Press, ISBN-13: 9781509302581
4. **Mark J. Price** — ASP.NET Core in Action, Manning Publications, ISBN-13: 9781617294617
5. **Jon P Smith** — Entity Framework Core in Action (2nd Edition), Manning Publications, ISBN-13: 9781617298363
6. **Microsoft Documentation** — ASP.NET Core & Entity Framework Core Official Documentation, Microsoft Learn
<https://learn.microsoft.com/aspnet/core>
<https://learn.microsoft.com/ef/core>

Course Code: CA-326(B)
Course Title: Data Analytics – II

Course Code: CA-326 (B)	Course Category: DSE-3
Course Title: Data Analytics - II (Power BI)	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To introduce students to data analytics, business intelligence, and the Power BI ecosystem. 2. To enable students to import, clean, transform, and combine data from multiple sources. 3. To develop skills in data modeling and basic DAX calculations for analysis. 4. To equip students to create effective visualizations and interactive dashboards in Power BI. 5. To familiarize students with advanced visualization features, including custom visuals and report formatting best practices.	
Course Outcomes: Students will be able to -	Cognitive Level
1. Understand the fundamentals of data analytics, business intelligence and the Power BI ecosystem.	2
2. Implement Import, clean, transform and combine data from multiple sources using Power BI and Power Query.	3
3. Apply data models, define relationships, and perform basic calculations using DAX functions.	3
4. Analyze and Design effective visualizations, including charts, tables, matrices, filters, and slicers.	4

Course Content:

Unit 1: Introduction to Data Analytics & Data Connection using Power BI (07 L, 10 M)

- Basics of Data Analytics and Business Intelligence
- Role of Visualization in Analytics
- Overview of Power BI Ecosystem (Power BI Desktop, Service, Mobile)
- Understanding Interface, Ribbons, Panels, and Panes
- Concepts: Datasets, Reports, Dashboards

Unit 2: Creating Visualization (08 L, 15 M)

- Importing data from multiple sources (Files, Databases)
- Query Editor (Power Query): Data Profiling & Transformation
- Cleaning data: Remove Duplicates, Replace Values, Format Types
- Merging and Appending Queries.
- Data modeling concepts: Star Schema, Relationships
- Introduction to Measures & Calculated Columns

Unit 3: Data Modeling & DAX Basics (08 L, 15 M)

- Basic DAX (Data Analysis Expressions):
- SUM, AVERAGE, COUNTROWS
- Hierarchies and role of keys
- Data Visualization:
 - Bar chart
 - Column chart
 - Line chart
 - Pie chart

- Table & Matrix
- Filters & Slicers

Unit 4: Data Visualization Techniques

(07 L, 10 M)

- Custom visuals & Gallery
- Sorting, filtering & slicers
- Conditional formatting & drill-down
- Designing interactive dashboards
- Sync slicers & bookmarks
- Drill-throughs & tooltips
- Themes & report formatting best practices

Reference Books:

- Beginning Microsoft Power BI: A Practical Guide to Self-Service Data Analytics – *Dan Clark*; Apress; 1st Edition (2020); ISBN-10: 1484267192, ISBN-13: 9781484267196
- Learn Power BI – Second Edition – *Greg Deckler*; Packt Publishing; 2nd Edition (2022); ISBN-13: 9781801811958
- Learning Microsoft Power BI: Transforming Data into Insights – *Jeremey Arnold*; Shroff/O'Reilly; 1st Edition; ISBN-13: 9789355422507
- Microsoft Power BI Cookbook – *Brett Powell*; Packt Publishing; (ISBN-13: 9781788297233)
- Microsoft Power BI Quick Start Guide – *Devin Knight*; Packt Publishing; 3rd Edition; ISBN-13: 9781789138221

Course Code: CA-326 (C)
Course Title: Machine Learning

Course Code: CA-326(C)	Course Category: DSE-3
Course Title: Machine Learning	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1.To introduce fundamental concepts of Machine Learning and data handling techniques. 2.To develop understanding of regression, classification, and clustering algorithms. 3.To enable students to apply Machine Learning techniques to real-world problems.	
Course Outcomes:	Cognitive Level
1. Explain basic Machine Learning concepts, datasets, and preprocessing techniques.	2
2. Apply regression and classification algorithms to solve simple problems.	3
3. Identify and implement suitable Machine Learning models for real-world applications.	3
4. Evaluate the performance of Machine Learning models using appropriate metrics and interpret the results effectively.	4

Course Content:

Unit 1 – Introduction to Machine Learning (8 L 15 M)

- Machine Learning (Features of Machine Learning, Applications of Machine Learning,
- Types of Machine Learning (Supervised Learning, Unsupervised Learning, Reinforcement Learning)
- What is a Dataset? Dataset Structure (Features, Labels, Instances)
- Types of Data (Structured and Unstructured Data, Numerical and Categorical Data, Training, Validation, and Test Data)
- Data Collection Methods (Survey, Sensors, APIs, Databases)
- Machine Learning Life Cycle (Problem Definition, Data Collection, Data Pre-processing, Model Selection, Model Training, Model Evaluation)
- Data Pre-processing (Data Cleaning Outliers, Feature Selection)

Unit 2 – Learning with Regression (8 L 15 M)

- Regression (What is Regression? Difference between Regression and Classification, Need and Use of Regression Analysis)
- Types of Linear Regression(Polynomial, Multiple Linear Regression)
- Linear Regression(Hypothesis Function, Assumptions of Linear Regression)
- Concept of Cost Function
- Model Evaluation(Mean Absolute Error (MAE),Mean Squared Error (MSE), R-Square

Unit 3 – Classification & Clustering Algorithms (8 L 15 M)

- Classification Algorithms
 - Logistic Regression
 - Binary Classification
 - K-Nearest Neighbour (KNN)
 - Naïve Bayes Classifier
 - Support Vector Machine (SVM)
- Clustering
 - Types of Clustering(Partitioning ,Hierarchical)

Unit 4 – Problem Definition & Real-World Applications

(6 L 15 M)

- Problem Definition Framework
- Understanding Business Problem
- Converting Business Problem to ML Problem
- Identifying Input (Features) and Output (Target)
- Selecting Appropriate Algorithm
- Steps in Machine Learning Problem Solving (Define Objective, Data Preparation, Model Selection, Model Training and Testing, Performance Evaluation)
- Evaluation Metrics (Accuracy, Precision, Recall, F1-Score, Confusion Matrix)
- Real-World Applications (Spam Detection (Email Filtering, Image & Video Recognition. Demand Forecasting (Sales Prediction), Virtual Personal Assistant, Recommendation Systems)

References:

1. **T. M. Mitchell**, *Machine Learning*. New York, NY, USA: McGraw-Hill, 1997.
2. **E. Alpaydin**, *Introduction to Machine Learning*, 4th ed. Cambridge, MA, USA: MIT Press, 2020.
3. **S. Marsland**, *Machine Learning: An Algorithmic Perspective*, 2nd ed. Boca Raton, FL, USA: CRC Press, 2015.
4. **C. M. Bishop**, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
5. **A. Géron**, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
6. **I. Goodfellow**, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

Course Code: CA-327(A)
Course Title: Practical based on Web Technology-II

Course Code: CA-327 (A)	Course Category: DSE - 4
Course Title : Practical based on web Technology-II (ASP.NET)	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understand the fundamentals of web application development using ASP.NET Core. 2. To learn MVC architecture and request–response processing in web applications. 3. To implement database connectivity using Entity Framework Core with Microsoft SQL Server. 4. To develop RESTful Web APIs and understand dependency injection, authentication, and authorization concepts. 5. To build industry-oriented web applications using ASP.NET Core.	
Course Outcomes: After completing this course, students will be able to:	Cognitive Level
1. Explain ASP.NET Core framework, web architecture, and MVC components.	2
2. Develop ASP.NET Core MVC applications using controllers, views, routing, and validation techniques.	3
3. Implement database connectivity and perform CRUD operations using Entity Framework Core with Microsoft SQL Server.	3
4. Develop RESTful Web APIs and apply dependency injection and authentication mechanisms in ASP.NET Core applications.	3

Course Content:

Assignments

Practical 1: Creation of ASP.NET Core MVC Application

Objective: To understand the structure of an ASP.NET Core MVC project.

Task & Expected Output: Create a new ASP.NET Core MVC application using Visual Studio, run the application to observe default output, and study Controllers, Models, Views, wwwroot, and appsettings.json files, resulting in a successfully running MVC application with default home page.

Practical 2: MVC Flow – Controller and View

Objective: To understand interaction between Controller and View.

Task & Expected Output: Create a HomeController with Index action method and a Razor View to display a welcome message, resulting in the message being displayed in the browser using proper MVC flow.

Practical 3: SQL Server Configuration with Entity Framework Core

Objective: To connect ASP.NET Core application with Microsoft SQL Server using EF Core.

Task & Expected Output: Install SQL Server (LocalDB/Express), install EF Core packages, configure connection string in appsettings.json, and verify successful database connection.

Practical 4: Model and DbContext Creation (Code First Approach)

Objective: To create model classes and configure database context.

Task & Expected Output: Create Product model with Id, Name, Price, Description, and Category properties, create ApplicationDbContext class, register DbContext in Program.cs, resulting in proper model configuration.

Practical 5: Migration and Database Creation

Objective: To create database using EF Core migrations.

Task & Expected Output: Execute Add-Migration and Update-Database commands to generate database schema in SQL Server, resulting in database and Product table creation.

Practical 6: Display Product List (Read Operation)

Objective: To retrieve and display data from SQL Server.

Task & Expected Output: Insert sample product records, fetch data using EF Core, and display products in Razor View in tabular format.

Practical 7: Add Product (Create Operation)

Objective: To insert data into SQL Server using EF Core.

Task & Expected Output: Create Add Product form, submit form using model binding, save product to database, resulting in product added successfully.

Practical 8: Update and Delete Product (Update & Delete Operations)

Objective: To perform Update and Delete operations using EF Core.

Task & Expected Output: Create Edit and Delete functionality, update product details, delete selected product, resulting in successful database modification.

Practical 9: One-to-Many Relationship Implementation

Objective: To implement relationship between two entities.

Task & Expected Output: Create Category and Product models with one-to-many relationship, apply migration, and display related data successfully.

Practical 10: LINQ Queries for Data Filtering

Objective: To apply LINQ queries in EF Core.

Task & Expected Output: Implement filtering using Where and Select methods to display filtered product results.

Practical 11: Form Validation using Data Annotations

Objective: To implement client-side and server-side validation.

Task & Expected Output: Apply Required, StringLength, and Range attributes in Product model, display validation messages during form submission.

Practical 12: Basic Authentication (User Registration & Login)

Objective: To implement basic authentication mechanism.

Task & Expected Output: Create User model, implement registration and login functionality, validate credentials, and redirect to secure page upon successful login.

Practical 13: Dependency Injection and Error Handling

Objective: To understand service injection and exception handling.

Task & Expected Output: Create service class, inject service into controller using dependency injection, handle exceptions using try-catch and custom error pages.

Practical 14: RESTful Web API for Product CRUD

Objective: To develop RESTful Web API using ASP.NET Core.

Task & Expected Output: Create API Controller, implement GET, POST methods using EF Core with SQL Server, return JSON responses, and test API using Swagger/Postman.

Tools & Software Required

- Visual Studio 2022 or above
- .NET 6 / .NET 7 / .NET 8 SDK
- Microsoft SQL Server (Express Edition / LocalDB)
- SQL Server Management Studio (SSMS)

- Web Browser (Chrome / Edge / Firefox)
- Postman (optional, for API testing)
- Entity Framework Core Tools (via NuGet / .NET CLI)

Course Code: CA-327(B)
Course Title: Practical based on Data Analytics-II

Course Code: CA-327 (B)	Course Category: DSE-4
Course Title: : Practical based on Data Analytics-II (Power BI)	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To understanding of data analysis concepts using Microsoft Power BI. 2. To enable students to perform data transformation, modelling, and visualization using Power BI tools. 3. To equip students with the skills required to design and develop interactive reports and dashboards that support effective data-driven decision-making.	
Course Outcomes: Students will be able to -	Cognitive Level
1. Understand and apply the process of importing, transforming, and modeling datasets using Microsoft Power BI and Power Query Editor.	2
2. Apply calculated columns and measures, and use basic DAX functions to analyze data for meaningful insights.	3
3. Analyze, design, and publish interactive dashboards with visuals and slicers to support business decision-making.	4

Course Content:

Assignments:

1. Download or create a simple dataset in Excel/CSV format (e.g., Sales Data, Student Performance Data and Product Data).
2. Connect the dataset to Microsoft Power BI Desktop
3. Load the data into Power BI.
4. Using Power Query Editor for transformations
5. Creating relationships between tables
6. Creating calculated columns and measures
7. Use of simple DAX functions.
8. Designing visual reports
9. Interacting with visuals using slicers
10. Building and publishing dashboard
11. Development of Interactive Visual Dashboards using dataset.

Course Code: CA-327(C)
Course Title: Practical on Machine Learning

Course Code: CA-327(C)	Course Category: DSE-4
Course Title: Practical on Machine Learning	Type: Practical
Total Contact Hours: 60 (4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Understand fundamental concepts of Machine Learning and data preprocessing techniques. 2. Apply regression algorithms and evaluate their performance using appropriate metrics. 3. Implement classification and clustering algorithms for solving real-world problems. 4. Analyze and interpret model results using performance evaluation measures and visualizations.	
Course Outcomes:	Cognitive Level
1. Understand Machine Learning concepts, dataset structure, and preprocessing techniques.	2
2. Apply regression algorithms and compute performance.	3
3. Implement classification algorithms and evaluate using accuracy measures.	3
4. Implement clustering algorithms and interpret clustering results.	3

Course Content:

Prerequisite Knowledge

• **Basic Mathematics & Statistics:**

Knowledge of algebra, matrices, mean, median, variance, standard deviation, correlation, and basic probability concepts.

• **Basic Computer Skills:**

Ability to install software, run Python programs, work with CSV files, and use basic command-line operations.

1. Write a program to study and demonstrate the use of Machine Learning libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn.
2. Write a program to load a dataset and identify its features, labels, and number of instances.
3. Write a program to perform Exploratory Data Analysis (EDA) on a given dataset using statistical summaries and visualizations.
4. Write a program to handle missing values and detect/remove outliers from a dataset.
5. Write a program to perform feature scaling using Normalization and Standardization techniques.
6. Write a program to split a dataset into Training, Validation, and Test sets.
7. Write a program to implement Simple Linear Regression and visualize the regression line.
8. Write a program to implement Multiple Linear Regression on a given dataset.
9. Write a program to implement Polynomial Regression and compare it with Linear Regression.
10. Write a program to calculate and compare MAE, MSE, and R^2 score for regression models.
11. Write a program to compute and visualize the Cost Function (Mean Squared Error) for Linear Regression.
12. Write a program to implement Logistic Regression for Binary Classification.
13. Write a program to implement the K-Nearest Neighbour (KNN) classifier and evaluate its performance.
14. Write a program to implement the Naïve Bayes classifier on a given dataset.
15. Write a program to implement the Support Vector Machine (SVM) classifier.
16. Write a program to implement K-Means Clustering and visualize the clusters.

Required Tools

- Jupyter Notebook/Google Colab/VS Code / PyCharm

Course Code: CA-328(A)
Course Title: Social Media Marketing

Course Code: Min 08	Course Category: Minor
Course Title: Social Media Marketing	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To explain the concepts, evolution, and significance of social media marketing. 2. To examine major social media platforms and their business applications. 3. To apply content creation and advertising strategies for online campaigns. 4. To analyze campaign performance using social media analytics. 5. To design a basic social media marketing plan for a product or service.	
Course Outcomes: After completion of the course, the student will be able to	Cognitive Level
1. Describe the role of social media in digital marketing and business communication.	2
2. Identify suitable social media platforms for different business objectives.	2
3. Develop structured content and promotional strategies for online engagement.	3
4. Interpret performance metrics and campaign insights.	4
5. Prepare a simple and effective social media marketing plan.	6

Course Content:

Unit 1: Foundations of Social Media Marketing

(08 L, 10 Marks)

- Introduction to Digital Marketing
- Evolution and growth of Social Media Marketing
- Traditional Marketing vs Social Media Marketing
- Digital consumer behavior
- Role of social media in business and IT sector
- Social media marketing process and strategy overview

Unit 2: Social Media Platforms and Content Strategy

(08 L, 15 Marks)

- Business Applications and Features of:
 - Facebook
 - WhatsApp
 - Instagram
 - LinkedIn
 - YouTube
- Creating and optimizing business profiles
- Types of content: posts, stories, reels, videos, live sessions
- Content planning and preparation of editorial calendar
- Hashtag strategy and techniques for audience engagement

Unit III: Social Media Advertising and Analytics

(08 L, 15 Marks)

- Organic marketing vs paid marketing
- Advertising models and campaign objectives
- Budget planning and audience targeting strategies
- Introduction to advertisement management tools
- Key Performance Indicators (KPIs)
- Basics of analytics and performance measurement
- Interpretation of insights and analytical reports

Unit IV: Strategy Development, Ethics, and Legal Aspects (06 L, 10 Marks)

- Steps in developing a social media marketing plan
- Brand positioning and digital storytelling
- Fundamentals of influencer marketing
- Online reputation management
- Ethical issues in social media marketing
- Data privacy and cyber laws related to digital promotion

References:

1. Ryan, Damian & Jones, Calvin. Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation. Kogan Page India Pvt. Ltd., New Delhi, 2019.
2. Gupta, Seema. Digital Marketing. McGraw Hill Education (India) Pvt. Ltd., New Delhi, 2018.
3. Kapoor, Neha. Digital Marketing for Beginners. BPB Publications, New Delhi, 2021.
4. Bansal, Ruchi & Bansal, Saurabh. Social Media Marketing. Taxmann Publications Pvt. Ltd., New Delhi, 2022.
5. Mishra, Dr. Nishikant & Mahalik, Dr. Debashis. Digital Marketing: Theory and Practice. Himalaya Publishing House, Mumbai, 2020.
6. Jain, Vandana & Yadav, Anshul. E-Marketing and Digital Marketing. Pearson India Education Services Pvt. Ltd., 2019.

Course Code: CA-328(B)
Course Title: Linear Algebra-II

Course Code: CA-328(B)	Course Category: Minor-08
Course Title: Linear Algebra-II	Type: Theory
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Apply advanced vector space concepts including inner products and orthogonality in problem solving. 2. Apply linear transformations, matrix decompositions, and orthogonalization techniques. 3. Analyze least squares methods, quadratic forms, and PCA for data modeling and optimization. 4. Analyze spectral methods and matrix-based techniques used in differential equations, data science, and AI applications.	
Course Outcomes: After completion of the course, students will be able to:	Cognitive Level
1. Apply inner product space concepts, Gram–Schmidt process, and orthogonal projections to construct orthonormal bases and analyze positive definite matrices.	3
2. Analyze linear transformations using kernel, range, rank–nullity theorem, diagonalization, orthogonal matrices, and basic SVD concepts.	4
3. Apply least squares approximation, normal equations, quadratic forms, and covariance matrices to solve optimization and data fitting problems.	3
4. Analyze eigenvalue-based methods in spectral techniques, matrix exponentials, PageRank concept, coding theory, cryptography, and machine learning models.	4

Course Content:

UNIT 1 – Advanced Vector Spaces & Inner Product Spaces

(7 L, 12 M)

- Review of vector spaces
- Linear independence and span
- Inner product spaces
- Orthogonality
- Gram–Schmidt process
- Orthogonal projections
- Positive definite matrices

UNIT 2 – Linear Transformations & Decompositions

(8 L, 13 M)

- Linear transformations between vector spaces
- Kernel and Range
- Rank–Nullity Theorem (conceptual)
- Diagonalization of symmetric matrices
- Matrix powers
- Introduction to Singular Value Decomposition (conceptual)
- Orthogonal matrices
- Applications in data representation

UNIT 3 – Least Squares, Optimization & PCA

(8 L, 13 M)

- Least squares approximation
- Normal equations
- Curve fitting
- Quadratic forms
- Positive definite matrices in optimization
- Covariance matrix
- Principal Component Analysis (PCA – geometric intuition)
- Data compression concept

UNIT 4 – Spectral Methods & AI Applications

(7 L, 12 M)

- Matrix exponential (basic idea)
- Systems of linear differential equations (conceptual)
- Eigenvalue applications in recurrences
- PageRank concept (spectral idea)
- Error correcting codes (basic idea)
- Cryptography using matrices (mod arithmetic)
- Machine Learning connection (Regression, Classification – matrix view)

Reference Book:

1. **P. Selinger**, Matrix Theory and Linear Algebra, open textbook, Dalhousie University, Halifax, NS, Canada.
2. <https://www.mathstat.dal.ca/~selinger/linear-algebra/downloads/LinearAlgebra.pdf>.
3. Linear Algebra and Its Applications – **David C. Lay, Steven R. Lay & Judi J. McDonald**; Pearson Education; 5th Edition (2015); ISBN-13: 9780321982384.
4. Introduction to Linear Algebra – **Gilbert Strang**; Wellesley-Cambridge Press; 5th Edition (2016); ISBN-13: 9780980232776.
5. Advanced Engineering Mathematics – **Erwin Kreyszig**; Wiley; 10th Edition (2011); ISBN-13: 9780470458365.
6. Matrix Analysis – **Roger A. Horn & Charles R. Johnson**; Cambridge University Press; 2nd Edition (2013); ISBN-13: 9780521548236.
7. Linear Algebra with Applications – **Steven J. Leon**; Pearson Education; 9th Edition (2014); ISBN-13: 9780321962218.
8. Applied Linear Algebra – **Ben Noble & James W. Daniel**; Pearson Education; 3rd Edition (1988); ISBN-13: 9780130412942.

Course Code: CA-329(A)
Course Title: Practical based on Social Media Marketing

Course Code: CA-329(A)	Course Category: MIN-09
Course Title: Practical based on Social Media Marketing	Type: Practical
Total Contact Hours: 30 (2/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. To develop practical skills in creating and managing business profiles on social media platforms. 2. To enable students to design and publish engaging digital content aligned with business goals. 3. To equip learners with planning skills through editorial calendars and content strategies. 4. To introduce basic concepts of social media marketing, advertising and performance analysis.	
Course Outcomes:	Cognitive Level
1. Create and manage a professional business profile on social media platforms.	3
2. Design and publish relevant content (posts, reels and stories) for engagement.	6
3. Develop a structured editorial calendar for content planning.	4
4. Understand social media performance using basic analytics.	2

Course Content:

Practical 1: Create the Business Profile

- Create the business account on any one platform (Instagram / Facebook / WhatsApp)
- Choose the hypothetical business and includes Profile photo (logo), Bio/description (clear and concise), Contact details and Business category
- **Journal File:** Social media platform screenshots of business profile and brief explanation of business.

Practical 2: Content Creation and Posting

- Create and publish at least 2 posts (image + caption), 1 short video / reel and 1 story.
- Content should align with the business created in Practical 1.
- **Journal File:** Screenshots or links and brief explanation of content purpose

Practical 3: Editorial Calendar Design (To do plan)

- Prepare the **7-day content calendar** for a business include Date, Platform, Content type (post/reel/story), Topic, Caption idea
- **Journal File:** Table (Excel or Word format)

Practical 4: Hashtag and Engagement Strategy

- Select the niche – Business Style (e.g., fitness, education, food etc.)

- List out 10 relevant hashtags
- Write 2 sample captions using hashtags
- **Journal File:** Document with hashtags and captions with specific business type.

Practical 5: Social Media Advertising Plan

- Design a simple ad campaign includes Objective, Target audience, Budget, social media platform
- Create a sample ad (image + text)
- **Journal File:** Campaign plan document and Ad creative (image + caption)

Practical 6: Analytics and Performance Review

- Assume data for at least 2 posts includes Likes, Comments and Shares
- Identify best performing post and reason for performance
- **Journal File:** Table of data and short analysis report (1–2 pages)

Reference Text:

7. Ryan, Damian & Jones, Calvin. Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation. Kogan Page India Pvt. Ltd., New Delhi, 2019.
8. Gupta, Seema. Digital Marketing. McGraw Hill Education (India) Pvt. Ltd., New Delhi, 2018.
9. Kapoor, Neha. Digital Marketing for Beginners. BPB Publications, New Delhi, 2021.
10. Bansal, Ruchi & Bansal, Saurabh. Social Media Marketing. Taxmann Publications Pvt. Ltd., New Delhi, 2022.
11. Mishra, Dr. Nishikant & Mahalik, Dr. Debashis. Digital Marketing: Theory and Practice. Himalaya Publishing House, Mumbai, 2020.
12. Jain, Vandana & Yadav, Anshul. E-Marketing and Digital Marketing. Pearson India Education Services Pvt. Ltd., 2019.

Assessment Criteria:

Marking Scheme – College Assessment (CA): Total 20 Marks

- Viva-Voce: 10 Marks
- Journal File: 10 Marks

Marking Scheme - University Assessment (UA): Total 30 Marks

- Viva-Voce: 20 Marks
- Journal File: 10 Marks

Course Code: CA-329(B)
Course Title: Practical Linear Algebra-II

Course Code: CA-329(B)	Course Category: Minor-09
Course Title: Practical Linear Algebra-II	Type: Practical
Total Contact Hours: 60(4/week)	Course Credits: 02
College Assessment (CA) Marks: 20 Marks	University Assessment (UA): 30 Marks
Course Objectives: 1. Apply advanced linear algebra concepts through computational methods. 2. Implement matrix decompositions and dimensionality reduction techniques programmatically. 3. Analyze spectral methods and matrix properties using algorithms. 4. Develop computational models for applications in data science, security, and optimization.	
Course Outcomes: After completion of the course, students will be able to:	Cognitive Level
1. Implement Gram–Schmidt process, diagonalization check, and verification of Rank–Nullity theorem using programs.	3
2. Develop programs for least squares, covariance matrix computation	3
3. Apply matrix-based algorithms to real-world problems	4
4. Analyze matrix properties including positive definiteness using computational techniques.	4

Prerequisites for Linear Algebra

Students should have basic knowledge of:

- Mathematics: Basic algebra, coordinate geometry, matrices & determinants
- Programming: Variables, loops, conditions, functions
- Python Basics: Syntax, indentation, lists/arrays, importing libraries
- Basic Math: Vectors, matrices, linear equations, matrix multiplication

Software Tools Required

- Programming Environment: Python 3.x
- Libraries: NumPy, Matplotlib, Pandas, Scikit-learn, Pillow (PIL)

Practical Assignments:

1. Write a program for Gram–Schmidt orthogonalization process.
2. Write a program for checking matrix diagonalization.
3. Write a program for solving least squares problems.
4. Write a program for computing covariance matrix and performing PCA.
5. Write a program for performing Singular Value Decomposition (SVD).
6. Write a program for verifying Rank–Nullity theorem.
7. Write a program for implementing PageRank algorithm.
8. Write a program for compressing an image using SVD.
9. Write a program for implementing Hill Cipher using matrices.
10. Write a program for checking whether a matrix is positive definite.